

Dijkstra's Algorithm

Dijkstra's Algorithm is used to find the shortest path from a source node to all other nodes in a weighted graph where edge weights are non-negative.

Key Idea

The algorithm repeatedly selects the node with the smallest known distance from the source and updates the distances of its neighboring nodes.

Main Steps

1. Initialize distances

- Distance to the source node = 0
- Distance to all other nodes = ∞ (infinity)

2. Use a Min-Priority Queue

- Store nodes based on their current shortest distance.

3. Select the Node with Minimum Distance

- Remove the node with the smallest distance from the queue.

4. Relax Edges

- For each neighbor, check if a shorter path exists through the current node.
- If yes, update the distance.

5. Repeat

- Continue until all nodes are processed or the destination node is reached.

Example Application

If nodes represent cities and edges represent road distances, the algorithm finds the shortest route from one city to all other cities.

Real-World Uses

- Network routing protocols
 - OSPF (Open Shortest Path First)
 - IS-IS (Intermediate System to Intermediate System)
- GPS navigation systems
- Internet packet routing
- Logistics and transportation planning

Important Data Structure

The algorithm uses a Min-Priority Queue (often implemented using a heap) to efficiently choose the next node with the smallest distance.

Time Complexity

- Using a binary heap:
- $O((V + E) \log V)$
- where:
 - V = number of vertices
 - E = number of edges